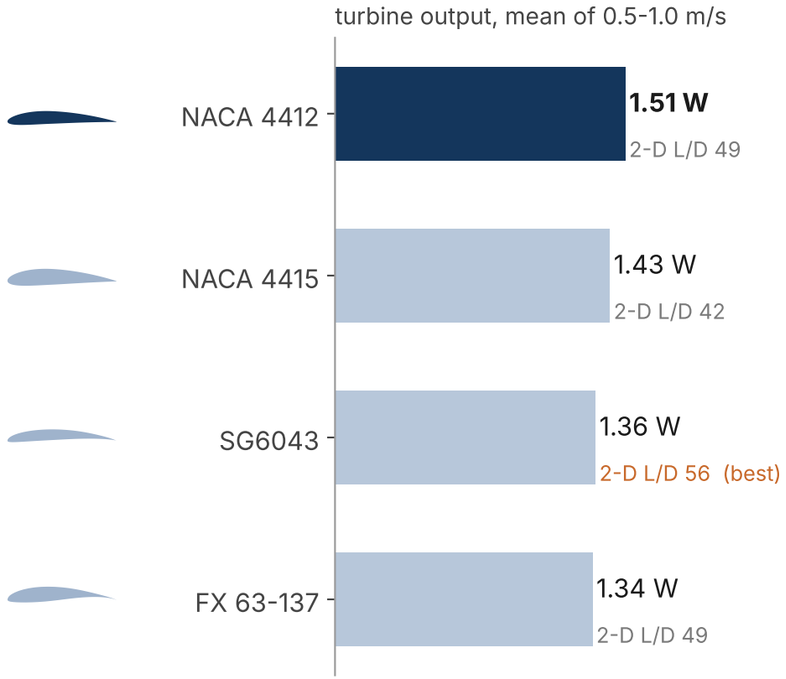
Iteration review: foil choice, design history, generator check

8-in hydrokinetic turbine for 0.5-1.0 m/s with the CubeMars GL60 KV25. Model predictions; nothing measured yet.

1 Foil choice: NACA 4412

The best 2-D foil (SG6043) is not the best turbine: NACA 4412 makes 11 % more power

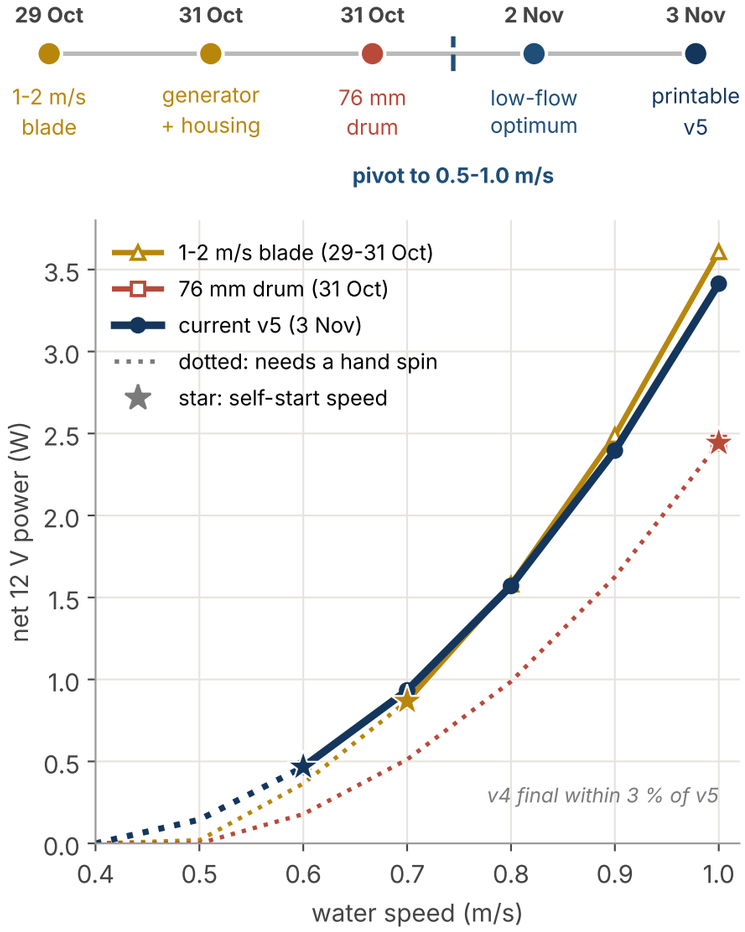


How it was chosen

- 7 open-source foils screened (NeuralFoil, an XFOIL surrogate); E387 and E818 dropped (low confidence)
- each foil built into rotors and scored as a whole turbine with the GL60
- NACA 4418 (18 % thick) used at the root to print

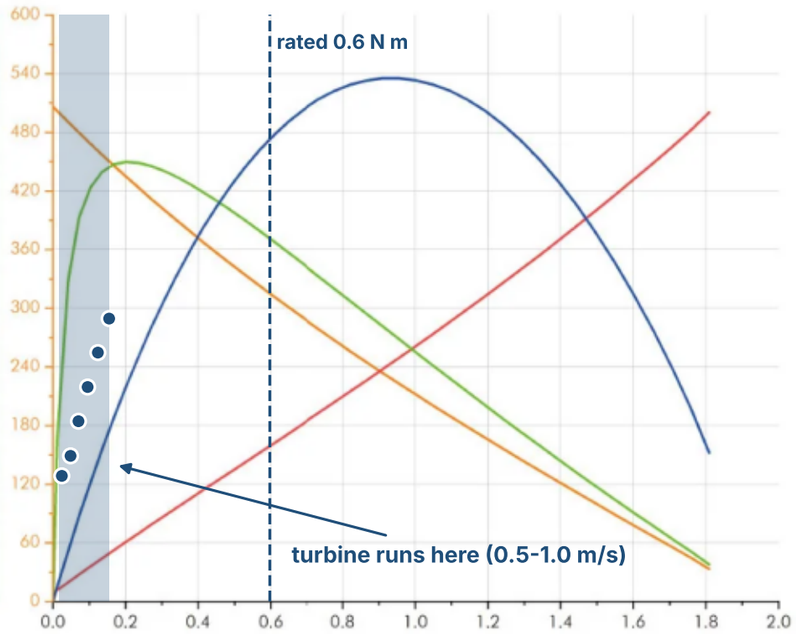
2 Design history

Re-targeting to 0.5-1.0 m/s: self-starts at 0.6 m/s (was 0.7-1.0) and +28 % power at 0.6 m/s



3 Generator: GL60 KV25

The turbine uses only 4-26 % of its rated torque; the motor's own drag is the open risk



CubeMars GL60 KV25 @ 24 VDC chart (supplied); lines:
speed (rpm, axis shown) current efficiency output power

OK	speed constant 21.5 rpm/V (516 rpm at 24 V; nameplate 25)
OK	torque per amp 0.44-0.48 N m/A, model 0.444
RISK	no-load draw up to 2.2 W at 516 rpm (model 0.32 W); if it is all motor drag, 0.7 m/s output falls 0.93 → 0.67 W

Next bench-test the GL60 (no-load drag, cogging) → print the rotor → flume test at 0.5-1.0 m/s → refit the model; revisit diameter if motor losses are high